

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Centralized eBay Customer Communication & Helpdesk Management System

Prepared in accordance with IEEE Std 830-1998 (Recommended Practice for SRS)

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0.5	2026-05-30	Solutions Architect	Added technical stack, NFRs, security section
0.9	2026-06-05	Operations Manager	Reviewed workflows, permissions, and SLAs
1.0	2026-06-12	Senior Business Analyst	Approved baseline for development handoff

Approvals

Role	Name	Responsibility	Status
Product Owner	[To be assigned]	Final scope approval	Approved
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1. Introduction

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the Centralized eBay Customer Communication & Helpdesk Management System (hereafter referred to as the "System" or the "Platform"). The document is prepared in accordance with IEEE Std 830-1998 and is intended to serve as the authoritative reference for all stakeholders including business owners, product managers, software architects, developers, quality assurance engineers, security officers, and operations staff.

The Platform addresses the operational pain point of managing customer communications scattered across multiple eBay seller accounts. By centralizing all incoming messages into a single role-aware web dashboard, the System will reduce response time, improve accountability through assignment workflows, and provide management with visibility into team productivity and customer-service performance.

This document is implementation-ready. Database schemas, REST/GraphQL API contracts, wireframes, and an iterative delivery plan can be derived directly from the requirements, rules, workflows, and acceptance criteria specified herein.

2. Purpose

The purpose of this SRS is to provide an unambiguous and complete description of what the System shall do and shall not do. Specifically, this document aims to:

- Document the business problem and the operational objectives that justify the build.
- Specify all functional requirements grouped by capability area (FR-01 ... FR-NN).
- Specify all non-functional requirements covering performance, scalability, security, availability, maintainability, and usability (NFR-01 ... NFR-NN).
- Codify enterprise business rules (BR-01 ... BR-NN) that govern conversation lifecycle, assignment, categorization, and SLA behavior.
- Define role-based permissions, audit obligations, and security controls.
- Provide acceptance criteria so that delivery can be objectively measured.
- Serve as a contractual baseline between the business and the development team.

3. Scope

3.1 In Scope

- Centralized ingestion of customer messages from six eBay seller accounts (Main, Marine, Tech, Ship, Tools, Trade) with extensibility for additional accounts via configuration.
- Conversation-based threading where one eBay buyer username maps to one conversation per seller account.
- AI-based automatic categorization of incoming conversations and manual override capability.

- Role-based access control with three primary roles: Admin, Operation Manager, and User.
- Assignment and reassignment workflows driven by Admin and Operation Manager roles.
- Reply composition from within the System and bi-directional synchronization back to eBay.
- Template management (CRUD) for canned responses centrally controlled by Admins.
- In-app notification center with bell icon, unread indicator, and read management.
- Comprehensive audit logging covering user, conversation, message, authentication, and system events.
- User and Manager analytics dashboards with daily, weekly, and monthly metrics.
- Local persistence of messages and attachments to minimize repeated eBay API calls.
- Secure storage of eBay credentials and tokens using encryption at rest.

3.2 Out of Scope (Version 1.0)

- Customer-facing interfaces. The platform is strictly internal.
- Email, SMS, push, or webhook notifications. Notifications are in-app only.
- Escalation engine for overdue SLA conversations (visual indicators only).
- Integration with marketplaces other than eBay (e.g., Amazon, Shopify, Walmart).
- Public REST/GraphQL APIs for third-party consumers.
- Self-service customer portal, knowledge base, or chatbot front-end.
- Multi-language UI (English-only in v1.0; i18n-ready architecture required).
- Mobile native applications (responsive web only).

3.3 Business Objectives

- Reduce average first-response time to customer messages by at least 40%.
- Eliminate manual account-switching by support staff (target: zero direct logins to eBay seller back office for messaging).
- Increase team accountability through assignment, status tracking, and audit trails.
- Enable management to make data-driven staffing and workload decisions.
- Maintain a 24-hour SLA on at least 95% of customer conversations.

4. Definitions, Acronyms & Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification.
RBAC	Role-Based Access Control — permissions granted via roles, not directly to users.
SLA	Service Level Agreement — the time within which a conversation must receive a reply.
Conversation	A persistent thread of messages between the organization and one eBay buyer on

	one seller account.
Category	An AI-assigned classification of a conversation (e.g., Cancellation, Return Request).
Assignment	The act of designating a User as the owner of a specific conversation.
Template	A pre-written reply text body managed centrally by Admins.
Sync	The process of fetching new eBay messages and pushing outbound replies.
Unassigned Category Queue	A holding area visible to Admins/Managers for conversations the AI could not classify.
Audit Log	An immutable, append-only record of security-relevant or business-relevant events.
FR	Functional Requirement.
NFR	Non-Functional Requirement.
BR	Business Rule.
UC	Use Case.
UI / UX	User Interface / User Experience.
API	Application Programming Interface.
CRUD	Create, Read, Update, Delete operations.
CSRF	Cross-Site Request Forgery.
TLS	Transport Layer Security (HTTPS encryption).
JWT	JSON Web Token — used for stateless authentication.
ORM	Object-Relational Mapper (e.g., SQLAlchemy).

5. Product Overview

5.1 Product Perspective

The System is a new, internally hosted web application that integrates with the eBay Trading and Messaging APIs. It operates as the canonical source of truth for all customer communications across all connected eBay seller accounts. Inbound messages are synchronized into the System's local PostgreSQL store; outbound replies are persisted locally and delivered to eBay via the eBay API. The System does not replace eBay; it sits alongside it as a productivity layer for support staff.

5.2 Product Functions (High Level)

- Authenticated, role-based access to a unified conversation inbox.
- Periodic and manual synchronization with multiple eBay accounts.
- AI-driven automatic categorization with manual override.
- Conversation assignment, reassignment, and status lifecycle management.
- Composition, templating, and dispatch of replies to eBay buyers.
- Attachment ingestion, local storage, viewing, and download.
- Real-time in-app notifications.
- Operational and individual analytics dashboards.
- Comprehensive audit logging with filter, search, and export.
- Administrative management of users, roles, categories, templates, and eBay accounts.

5.3 Operating Environment

- Server-side: Linux (Ubuntu 22.04 LTS or equivalent), containerized via Docker, orchestrated via Docker Compose or Kubernetes.
- Client-side: Latest two major versions of Chrome, Edge, Firefox, and Safari. Minimum viewport width 1280 px (desktop-first responsive).
- Database: PostgreSQL 14 or higher.
- Background workers: Celery with Redis broker.
- External dependencies: eBay Trading/Messaging APIs; OpenAI-compatible LLM endpoint (or local model) for categorization.

6. Stakeholders

Stakeholder	Interest / Responsibility
Business Owner / Sponsor	Funds the initiative; expects ROI through faster response times and lower per-message labor cost.
Operations Manager	Day-to-day owner; needs workload visibility, reassignment power, and team analytics.

Support Staff (Users)	Primary end users; need a fast, intuitive UI to triage and reply efficiently.
System Administrator	Manages eBay accounts, users, categories, templates, and platform configuration.
Security Officer	Validates compliance with credential storage, audit, and access-control requirements.
Development Team	Implements and maintains the platform; consumes this SRS as the build specification.
QA Team	Validates implementation against the acceptance criteria in Section 20.
Finance / Procurement	Tracks API and infrastructure cost; benefits from caching to control eBay API usage.

7. User Roles

The System defines three primary roles. All permissions in the System are granted via role membership (RBAC). A user may belong to exactly one role at a time. Role changes are audited.

7.1 Admin

The Admin is the highest privileged role. Admins are responsible for platform configuration, user lifecycle management, integration management (eBay accounts), category and template governance, and access to audit logs. Admins inherit all Operation Manager permissions plus exclusive administrative rights.

7.2 Operation Manager

The Operation Manager is the operational owner of day-to-day support work. Managers supervise Users, balance workload, reassign conversations, modify category access, and review team analytics. Managers do not have platform-administration rights such as adding eBay accounts, deleting users, or modifying global configuration.

7.3 User

Users are front-line support staff. Users can only view conversations that are (a) assigned to them directly, or (b) belong to categories assigned to them and not yet assigned to a specific colleague (where business rules permit visibility). Users can compose replies, change conversation status, mark spam, use templates, and view their own performance analytics.

8. Functional Requirements

Functional requirements are uniquely identified using the prefix FR-NN. Each requirement is atomic, verifiable, and traceable to a business objective or stakeholder need.

8.1 Authentication, Authorization & User Management

ID	Requirement	Actor(s)
FR-01	The System shall provide a secure login screen requiring username (or email) and password.	Admin / Manager / User
FR-02	The System shall hash all stored passwords using bcrypt (cost factor ≥ 12) or Argon2id. Plain-text passwords shall never be stored or logged.	All
FR-03	The System shall enforce a minimum password policy: 10+ characters, at least one uppercase letter, one lowercase letter, one digit, and one special character.	All
FR-04	The System shall lock an account for 15 minutes after 5 consecutive failed login attempts within 10 minutes.	All
FR-05	The System shall issue session tokens (e.g., JWT or server-side session) with idle expiration of 30 minutes and absolute expiration of 12 hours.	All
FR-06	The System shall provide a logout function that immediately invalidates the session token server-side.	All
FR-07	Admin shall be able to create new users specifying email, username, full name, role, and assigned categories.	Admin
FR-08	Admin and Operation Manager shall be able to edit user profile fields except role escalation to Admin (Admin only).	Admin / Manager
FR-09	Admin shall be able to disable a user (soft-state) preserving all historical data and audit links.	Admin / Manager (disable only)
FR-10	Admin shall be able to delete a user only after reassigning all conversations owned by that user.	Admin
FR-11	Admin shall be able to reset any user's password and trigger a forced password change at next login.	Admin
FR-12	The System shall enforce role-based access control on every API endpoint and UI route.	All

8.2 eBay Account Management

ID	Requirement	Actor(s)
FR-13	The System shall be pre-configured with six eBay accounts: Main, Marine, Tech, Ship, Tools, and Trade.	Admin
FR-14	Admin shall be able to add a new eBay account through a guided form capturing account label, eBay App credentials, OAuth token, refresh token, and token expiry.	Admin
FR-15	The System shall encrypt all eBay credentials and tokens at rest using AES-256-GCM with keys managed via an environment-isolated key vault.	System
FR-16	Admin shall be able to remove an eBay account; removal shall be soft-delete preserving historical conversations and audit trail.	Admin
FR-17	The System shall expose eBay account status: Active, Inactive, or Sync Error, with the timestamp and reason of last sync.	Admin / Manager
FR-18	The System shall automatically refresh OAuth tokens before expiry and log refresh events.	System
FR-19	Adding additional eBay accounts in the future shall not require code changes; the System shall load account configuration dynamically from the database.	Admin

8.3 Synchronization

ID	Requirement	Actor(s)
FR-20	The System shall poll the eBay Messaging API for new inbound messages on each connected account at a configurable interval (default: every 2 minutes).	System
FR-21	The System shall provide a manual Refresh Sync button in the header that triggers an immediate sync of the currently selected account(s).	All
FR-22	The System shall implement deduplication so that re-fetched messages do not create duplicate conversation entries.	System
FR-23	The System shall persist all messages, metadata, and attachments locally upon synchronization.	System
FR-24	Outbound replies shall be persisted locally and then dispatched to eBay; failed dispatches shall be retried with exponential backoff up to 5 attempts.	System
FR-25	The System shall maintain per-account sync state (last successful sync timestamp, last message ID, error count) and expose it to Admins.	Admin / Manager
FR-26	The System shall apply caching and rate limiting on outbound eBay API calls	System

	to remain well below eBay's per-day API quotas.	
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8.4 Conversations & Threading

ID	Requirement	Actor(s)
FR-27	The System shall maintain one conversation thread per (eBay account, eBay buyer username) tuple. Subsequent messages from the same buyer on the same account shall append to the existing conversation.	System
FR-28	The System shall preserve complete chronological history of every conversation, including original eBay message IDs, timestamps, sender, recipient, and message body.	System
FR-29	The System shall support the following conversation statuses: Open, Pending, Resolved, Reopened, Spam.	All
FR-30	New inbound messages shall create a conversation in Open status or, if the conversation existed previously and was Resolved, transition it to Reopened.	System
FR-31	Users, Managers, and Admins with access shall be able to change conversation status manually subject to BR-04.	User / Manager / Admin
FR-32	When a conversation is marked Spam, it shall be hidden from default views and routed to the Spam folder; the underlying messages remain audited.	User / Manager / Admin
FR-33	The System shall display SLA remaining time on each conversation row with color indicators (FR-58).	All
FR-34	Users shall be able to star/favorite a conversation to add it to their personal favorites list.	User / Manager / Admin

8.5 AI Categorization

ID	Requirement	Actor(s)
FR-35	The System shall automatically classify each new conversation into one of the configured categories using a Large Language Model (LLM) or equivalent classifier.	System
FR-36	The initial category set shall be: Cancellation, Return Request, Pre-Sales, Order Update, General Inquiry, Irrelevant.	Admin
FR-37	Admins shall be able to create, rename, and deactivate categories without code changes.	Admin
FR-38	The classifier shall return a confidence score; conversations below a configurable threshold (default 0.65) shall be routed to the Unassigned	System

	Category Queue.	
FR-39	The Unassigned Category Queue shall be visible only to Admins and Operation Managers.	Admin / Manager
FR-40	Users, Managers, and Admins shall be able to manually override the AI-assigned category at any time; overrides shall be audited.	All (per permissions)
FR-41	The System shall reclassify a conversation only when a new inbound message arrives; manually overridden categories shall be preserved unless overridden again by a privileged actor.	System

8.6 Assignment, Category Access & Routing

ID	Requirement	Actor(s)
FR-42	Admin and Operation Manager shall be able to assign any conversation to any active User.	Admin / Manager
FR-43	Admin and Operation Manager shall be able to reassign a conversation to a different User; the prior assignment shall be retained in audit history.	Admin / Manager
FR-44	The System shall enforce that Users see only conversations belonging to categories assigned to them, plus any conversation explicitly assigned to them irrespective of category.	System
FR-45	Admin shall be able to assign or remove category access for each User.	Admin
FR-46	Operation Manager shall be able to modify category assignments for Users within their team.	Manager
FR-47	Manual assignments shall override the default category restriction for that conversation only.	System

8.7 Reply Composition & Templates

ID	Requirement	Actor(s)
FR-48	Users with access to a conversation shall be able to compose and send replies from within the System.	User / Manager / Admin
FR-49	The reply composer shall support plain text with line breaks. Rich-text formatting is out of scope for v1.0.	System
FR-50	The reply composer shall provide a Template Picker allowing the User to insert and optionally edit a template body before sending.	User / Manager / Admin

FR-51	Outbound replies shall be sent to eBay via the eBay Messaging API and shall appear in the eBay seller account messaging UI.	System
FR-52	Admin shall be able to create, edit, and delete templates. Templates shall include a title, body, optional category tag, and last-modified metadata.	Admin
FR-53	Users and Managers shall be able to view and use templates but shall not be able to create, edit, or delete them.	System
FR-54	The System shall record every reply send event (success/failure) in the audit log including the message ID returned by eBay.	System
FR-55	The system shall validate outgoing messages before they are sent to eBay. The validation engine shall detect: • Prohibited words • Offensive language • Hate speech • Threatening language • Harassment • Scam-related content • Attempts to move communication off eBay • Sharing of personal contact information Examples: Call me at +91XXXXXXXXXX Email me at abc@gmail.com Let's complete the deal outside eBay	System

8.8 Attachments

ID	Requirement	Actor(s)
FR-56	The System shall ingest and locally store all attachments received from eBay during synchronization.	System
FR-57	Supported attachment formats in v1.0 are: JPG, JPEG, PNG, GIF, WEBP, HEIC, BMP. Other formats shall be stored as opaque binaries with a generic preview placeholder.	System
FR-58	Users with access to a conversation shall be able to view attachments inline (image preview) and download them. Download events shall be audited.	User / Manager / Admin

8.9 SLA, Filtering, Search & Sorting

ID	Requirement	Actor(s)
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FR-59	The System shall apply a standard SLA of 24 hours to every conversation, measured from the timestamp of the latest unanswered customer message. Indicators: Green (>50% remaining), Yellow (≤50% remaining), Red (≤10% remaining), Overdue (>24h).	System
FR-60	The System shall provide a multi-select account dropdown allowing Users to view conversations from all accounts, one account, or any subset.	All
FR-61	The System shall allow filtering by status, category, assigned user, account, SLA bucket, and date range.	All
FR-62	The System shall allow sorting by Last Activity Date, SLA Remaining Time, Subject, Account, and Status.	All
FR-63	The System shall support free-text search across customer username and message content. Order numbers embedded in message bodies shall be matched.	All
FR-64	Search shall be indexed (e.g., PostgreSQL full-text search or equivalent) to return results within 2 seconds for the production dataset (Section 9).	System

8.50 Notifications

ID	Requirement	Actor(s)
FR-65	Notifications shall be in-app only. No email, SMS, or push notifications are required in v1.0.	System
FR-66	The System shall generate notifications for: new conversation assignment, conversation reassignment, new customer reply on a conversation owned by the recipient, mention (if implemented), and conversation reopened.	System
FR-67	The header shall display a bell icon with a red dot when one or more unread notifications exist.	All
FR-68	Clicking the bell shall open a notification panel showing the latest 5 notifications with timestamp, summary, and a link to the relevant conversation.	All
FR-69	Users shall be able to mark notifications as read individually or as a group ('Mark all as read').	All
FR-70	Notifications shall be persisted in the database for at least 90 days.	System

8.11 Analytics

ID	Requirement	Actor(s)
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FR-71	The System shall provide a personal analytics view for Users showing: Assigned Today, Resolved Today, Pending Count, Total Handled Today, Weekly Count, and Monthly Count.	User
FR-72	The System shall provide a Manager analytics view showing: conversations received today/week/month, category-wise distribution, account-wise distribution, user-wise productivity, resolution metrics, and pending workload metrics.	Manager / Admin
FR-73	All analytics widgets shall support a date range selector and shall be exportable to CSV.	Manager / Admin
FR-74	Analytics queries shall not block the operational dashboard and shall use read-optimized data sources (materialized views or asynchronous aggregation).	System

8.12 Audit Logging

ID	Requirement	Actor(s)
FR-75	The System shall record an immutable audit entry for every action listed in Section 16.2 (User, Conversation, Message, Authentication, and System logs).	System
FR-76	Each audit entry shall capture: actor user ID, action code, timestamp (UTC), IP address, geographic location (resolved via GeoIP), affected entity type and ID, previous value, and new value.	System
FR-77	Audit logs shall be append-only; no role shall be able to delete or modify audit records.	System
FR-78	Admins shall be able to filter audit logs by actor, action, date range, entity, and IP, and shall be able to export filtered results to CSV.	Admin
FR-79	Audit logs shall be retained for a minimum of 24 months and shall be backed up daily.	System

9. Non-Functional Requirements

Non-functional requirements are identified using the prefix NFR-NN.

9.1 Performance

ID	Requirement
NFR-01	The Conversation List view shall load the first 50 records within 2 seconds at the 95th percentile under expected load.
NFR-02	Search queries shall return results within 2 seconds at the 95th percentile for a corpus of up to 250,000 messages.
NFR-03	Reply send (System → eBay) shall complete within 5 seconds at the 95th percentile, excluding eBay-side latency.
NFR-04	Background categorization shall classify a new conversation within 30 seconds of ingestion.
NFR-05	The manual Refresh Sync action shall return feedback to the user within 3 seconds (sync may continue in the background).

9.2 Scalability & Capacity

ID	Requirement
NFR-06	The System shall comfortably support a sustained inbound rate of 300–400 messages per day (vs. current 200/day) with no architectural changes.
NFR-07	The System shall support at least 25 concurrent authenticated users without degradation of NFR-01 through NFR-03.
NFR-08	Database storage shall be sized to retain 24 months of messages and attachments online; older data may be archived to cold storage.
NFR-09	The architecture shall permit horizontal scaling of FastAPI workers and Celery workers behind a load balancer.

9.3 Availability & Reliability

ID	Requirement
NFR-10	The platform shall target 99.5% monthly availability during business hours (excluding scheduled maintenance windows announced ≥72 hours in advance).
NFR-11	The System shall gracefully degrade when the eBay API is unavailable: existing data remains viewable; outbound replies queue for retry; a banner alerts users.

NFR-12	All Celery tasks shall be idempotent and resumable so that a worker crash does not corrupt data.
NFR-13	Database backups shall be taken daily with at least a 30-day retention and tested restoration quarterly.

9.4 Security

ID	Requirement
NFR-14	All client-server communication shall use TLS 1.2 or higher. Plain HTTP shall be disabled.
NFR-15	All credential and token storage shall be encrypted at rest using industry-standard symmetric encryption (AES-256).
NFR-16	The System shall implement CSRF protection on all state-changing endpoints (synchronizer tokens or SameSite=strict cookies).
NFR-17	Rate limiting shall be applied per IP and per user on authentication endpoints and reply submission endpoints.
NFR-18	All inputs shall be validated server-side using a typed schema (e.g., Pydantic); message bodies and template content shall be sanitized to prevent stored XSS.

9.5 Usability & Accessibility

ID	Requirement
NFR-19	The UI shall conform to WCAG 2.1 AA color contrast and keyboard navigation guidelines.
NFR-20	All destructive actions (delete user, remove eBay account, mark spam in bulk) shall require an explicit confirmation step.
NFR-21	Error messages shall be clear, actionable, and shall never expose stack traces or internal identifiers to end users.

9.6 Maintainability & Observability

ID	Requirement
NFR-22	The codebase shall maintain ≥80% unit test coverage on backend business logic.
NFR-23	The System shall expose structured JSON application logs and Prometheus-compatible metrics.
NFR-24	Database schema changes shall be managed exclusively via Alembic migrations; no manual schema edits in production.
NFR-25	Configuration shall be loaded from environment variables; no secrets shall be committed to source

	control.
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9.7 Portability & Deployment

ID	Requirement
NFR-26	The platform shall be deployable via Docker Compose for staging and Kubernetes (Helm chart) for production, without code changes.
NFR-27	All third-party services (LLM endpoint, GeoIP, eBay) shall be abstracted behind an interface to allow vendor substitution.

10. Business Rules

ID	Business Rule
BR-01	One eBay buyer username maps to exactly one conversation per eBay seller account. The same buyer messaging two different seller accounts produces two separate conversations.
BR-02	A conversation's complete chronological history shall never be truncated, edited, or deleted by any role.
BR-03	When a customer sends a new message to a Resolved conversation, the System shall automatically transition the conversation to Reopened (interpreted as Open for working purposes).
BR-04	Only the assigned User, the Operation Manager, and the Admin may change the status of a conversation. Status changes are audited.
BR-05	A User shall see a conversation if and only if (a) it is assigned to them, OR (b) it belongs to a category assigned to them and is unassigned. Manual assignments override category restrictions.
BR-06	AI-generated category assignments are non-binding; any privileged actor may override them. The override is the authoritative classification thereafter.
BR-07	Conversations with classifier confidence below the configured threshold are routed to the Unassigned Category Queue, visible only to Admin and Operation Manager.
BR-08	Spam conversations are excluded from default analytics counts (received, resolved, pending) unless explicitly opted into by the user via a filter.
BR-09	Templates are global. Users cannot personalize templates; if personalization is needed, Users may copy and edit the body in the reply composer before sending.
BR-10	Attachments downloaded by any user shall be logged with user ID, timestamp, and attachment ID.
BR-11	An eBay account cannot be hard-deleted while it has open conversations; it must be deactivated. Hard-deletion shall be supported only when no conversations remain (or for a documented data-retention purge).
BR-12	The 24-hour SLA clock pauses when conversation status is Pending (waiting for customer information) and resumes when a new customer reply arrives.
BR-13	Users cannot delete or edit messages once sent. Outbound replies are immutable after dispatch.
BR-14	A user account with assigned conversations cannot be deleted; conversations must be reassigned first.
BR-15	All times displayed in the UI shall be shown in the viewing user's local time zone; all stored times shall be UTC.

11. User Stories

User stories follow the standard format: As a <role>, I want <capability> so that <benefit>. Each story includes acceptance criteria.

US-01: Admin

As a Admin, I want to add a new eBay account through a guided form so that I can extend the platform's coverage without engineering involvement.

Acceptance Criteria:

- Form captures account label, eBay App ID, Cert ID, Dev ID, OAuth and refresh tokens.
- On save, the System validates the credentials against the eBay sandbox/production endpoint.
- On success, the account appears with Active status; on failure, an actionable error is shown.
- An audit entry 'eBay Account Added' is written.

US-02: Admin

As a Admin, I want to create users and assign categories to them so that support staff only see the conversation types they are authorized to handle.

Acceptance Criteria:

- User creation form captures name, email, username, initial password (or 'send invite'), role, and category list.
- Categories list is a multi-select.
- Created user receives a forced password change on first login.
- An audit entry 'User Created' is written including assigned role and categories.

US-03: Operation Manager

As a Operation Manager, I want to reassign a conversation from one user to another so that workload can be balanced when a team member is unavailable.

Acceptance Criteria:

- Reassignment is available from the conversation detail and from the list view (bulk).
- The receiving user gets an in-app notification within 5 seconds.
- Audit log entry 'Reassignment' records previous and new assignee.

US-04: User

As a User, I want to reply to a customer directly from the dashboard so that I avoid logging into eBay individually for each seller account.

Acceptance Criteria:

- Composer is available on the conversation detail view.
- Template picker is accessible from the composer.
- On send, reply appears in conversation thread within 2 seconds with status 'Sent'.
- Reply is delivered to eBay within 5 seconds (NFR-03); failures are retried automatically.

US-05: User

As a User, I want to mark irrelevant messages as Spam so that my inbox is uncluttered and metrics remain accurate.

Acceptance Criteria:

- Spam button is present on every conversation row and detail page.
- Confirmation prompt appears before status change.
- Spam conversations are hidden from default views and appear in the Spam folder.
- Action is audited with previous and new status.

US-06: User

As a User, I want to see only the conversation categories assigned to me so that I focus on my area of expertise and am not distracted.

Acceptance Criteria:

- List view automatically applies category filter to assigned categories.
- Conversations explicitly assigned to me appear regardless of category (per BR-05).
- Categories not assigned to me are not visible in any filter dropdown.

US-07: Operation Manager

As a Operation Manager, I want to view team analytics dashboards so that I can monitor productivity and identify bottlenecks.

Acceptance Criteria:

- Analytics page shows received today/week/month, category and account distributions, and per-user productivity.
- Date range selector adjusts all widgets.
- Export to CSV is available per widget.

US-08: Admin

As a Admin, I want to export audit logs filtered by user and date so that I can fulfill compliance and forensic requests.

Acceptance Criteria:

- Filter form supports actor, action, entity, date range, and IP.
- Export produces a CSV with all filtered fields.
- Export action itself is recorded in the audit log.

US-09: User

As a User, I want to receive an in-app notification when a conversation is assigned to me so that I can respond promptly without polling the dashboard.

Acceptance Criteria:

- Bell icon shows red dot within 5 seconds of assignment.
- Notification list shows the latest 5 with conversation subject and timestamp.

- Clicking the notification opens the conversation detail.

US-10: Admin

As a Admin, I want to create reusable reply templates so that the team responds consistently and quickly to common scenarios.

Acceptance Criteria:

- Template form captures title, body, and optional category tag.
- Users see templates in the composer template picker.
- Users cannot create, edit, or delete templates (FR-53).

12. Permission Matrix

Legend: ✓ = Full permission, P = Partial permission (see notes), X = No permission.

Capability	Admin	Operation Manager	User
Create users	✓	✓	X
Edit users	✓	P (cannot promote to Admin)	X
Disable users	✓	✓	X
Delete users	✓	X	X
Reset user passwords	✓	X	X
Assign categories to users	✓	✓	X
Remove category access	✓	✓	X
Assign conversations	✓	✓	X
Reassign conversations	✓	✓	X
View all conversations	✓	✓	X (own / category only)
View assigned conversations	✓	✓	✓
Reply to customers	✓	✓	✓ (own only)
Change conversation status	✓	✓	✓ (own only)
Mark conversations as spam	✓	✓	✓ (own only)
Use templates	✓	✓	✓
Create / edit / delete templates	✓	X	X
View team analytics	✓	✓	X
View personal analytics	✓	✓	✓
Add eBay accounts	✓	X	X
Remove eBay accounts	✓	X	X
Configure account settings	✓	X	X

View audit logs (all)	✓	P (operational only)	X
Manage system configuration	✓	X	X
View Unassigned Category Queue	✓	✓	X

13. Workflow Diagrams (Textual)

13.1 Inbound Message Workflow

1. Celery scheduler triggers eBay sync for each Active account (default every 2 minutes).
2. Worker calls eBay Messaging API and retrieves new messages since last_sync_cursor.
3. Worker normalizes each message and deduplicates against existing message IDs.
4. Worker resolves or creates a Conversation by (account_id, buyer_username).
5. If the Conversation is new OR was Resolved → status set to Open or Reopened (per BR-03).
6. Worker triggers the AI Categorization task with the message corpus of the conversation.
7. Categorization returns a category + confidence; below threshold → Unassigned Category Queue.
8. Notification rules (FR-65) fire for affected users (assigned user, or none if unassigned).
9. Conversation appears in the relevant list views; SLA clock starts (or resumes if previously paused).

13.2 Reply / Outbound Workflow

10. User composes reply in the conversation detail composer; optionally selects a template.
11. On send, frontend validates non-empty body and submits to /conversations/{id}/replies.
12. Backend authorizes the user against the conversation (BR-05).
13. Reply is persisted locally with status 'queued' and an outbound Celery task is dispatched.
14. Outbound worker calls eBay Messaging API; on success, status → 'sent' with eBay message ID stored.
15. On failure, exponential backoff retries up to 5 times; final failure flags the conversation with a 'Delivery Failed' banner and notifies the sender.
16. Audit log 'Reply Sent' is written including success/failure outcome.

13.3 Assignment Workflow

17. Admin or Operation Manager opens a conversation (or selects in bulk).
18. Picks 'Assign' or 'Reassign' and chooses a target User.
19. System validates target user is Active and not disabled.
20. Assignment record is created; previous assignment archived for audit.
21. In-app notification fires to the new assignee (FR-65).
22. Audit entry 'Assignment' or 'Reassignment' is written with prev/new values.

13.4 Status Lifecycle

Allowed transitions:

- Open → Pending (User awaits customer info; pauses SLA per BR-12).
- Open → Resolved (issue closed).
- Pending → Open (customer responded; SLA resumes).
- Pending → Resolved (manual close).
- Resolved → Reopened (auto, on new customer message — BR-03).
- Any → Spam (with confirmation).

- Spam → Open (un-spam by Manager/Admin).

13.5 Category Override Workflow

23. Authorized actor opens the conversation detail.
24. Selects 'Change Category' and picks the new category from the dropdown.
25. System updates the conversation category and writes an audit 'Category Change' entry with prev/new values.
26. If actor is a User and the new category is one they do not have access to, the System warns but proceeds (privileged actors only); for Users with no category-change rights on the target, the action is blocked.

13.6 eBay Account Onboarding Workflow

27. Admin clicks 'Add eBay Account' under eBay Account Management.
28. Wizard step 1 — Identification: enter Account Label, currency, time zone.
29. Wizard step 2 — Credentials: enter eBay App ID, Cert ID, Dev ID, and complete OAuth consent.
30. Wizard step 3 — Validation: System performs a test API call (GetUser or equivalent).
31. Wizard step 4 — Confirmation: account is saved with status Active and a first sync is queued.
32. Audit entry 'eBay Account Added' is written.

14. Dashboard Specifications

14.1 Top Header

Position	Component	Behavior
Left	Account Selector (multi-select)	Filters all conversations globally; selection persists per user.
Left	Refresh Sync Button	Triggers immediate sync for selected accounts; spinner shown until queued.
Center	System Name / Logo	Click navigates to the default Inbox view.
Right	Notification Bell	Red dot indicates unread; opens panel with latest 5 (FR-67).
Right	User Profile	Shows initials/avatar; menu with Profile and Preferences.
Right	Logout	Ends the session (FR-06).

14.2 Left Sidebar

Common to all roles:

- User information block (avatar, name, role).
- All Conversations
- Open
- Pending
- Resolved
- Reopened
- Spam

Admin and Operation Manager only:

- Unassigned Categories (queue)
- User Management
- Analytics
- Audit Logs (Admin full, Manager scoped)
- Template Management (Admin full, Manager read-only)
- eBay Account Management (Admin only)

14.3 Conversation List View Columns

Column	Description / Behavior
Checkbox	Multi-select for bulk actions (assign, reassign, mark spam, change status).
Star / Favorite	Toggle personal favorite (FR-34).

Category	AI or manually assigned; colored chip; clickable to filter.
Source	Fixed value 'eBay' in v1.0; designed for future marketplaces.
Account Name	eBay seller account label (Main, Marine, ...).
Customer Username	eBay buyer username; clickable to view conversation.
Subject	First N characters of the original eBay subject or first message.
Assigned User	Display name; '—' if unassigned.
Status	Open / Pending / Resolved / Reopened / Spam; colored badge.
SLA Remaining	Countdown with color (FR-58); 'Overdue' if >24h.
Last Activity Date	Timestamp of last message; sortable; default sort descending.

14.4 Conversation Detail View

- Header: Buyer username, account label, current category (editable), status (editable), SLA chip, assigned user (editable).
- Body: Chronological message thread with sender labels, timestamps, and attachment thumbnails.
- Composer: Plain text area with Template picker, character counter, and Send button.
- Side panel: Audit timeline (status changes, assignments, category changes), order numbers detected in messages, and quick-actions (Mark Spam, Resolve, Pending).

15. Analytics Specifications

15.1 User Analytics

Metric	Definition	Refresh
Assigned Today	Count of conversations newly assigned to the user where assignment timestamp is today.	≤ 60 s
Resolved Today	Count of conversations transitioned to Resolved by this user today.	≤ 60 s
Pending Count	Current count of conversations in Pending status owned by this user.	Realtime
Total Handled Today	Distinct count of conversations the user replied to or status-changed today.	≤ 60 s
Weekly Count	Total Handled across the current ISO week (Mon–Sun).	Hourly
Monthly Count	Total Handled across the current calendar month.	Hourly

15.2 Manager Analytics

Metric	Definition
Conversations Received (Today/Week/Month)	Count of inbound conversations created in the period.
Category-wise Distribution	Breakdown of conversations by AI/final category; pie or bar chart.
Account-wise Distribution	Breakdown of conversations by eBay seller account.
User-wise Productivity	Per-user assigned vs. resolved vs. pending; bar chart with rankings.
Resolution Metrics	Avg. time-to-first-reply, avg. time-to-resolution, % within SLA, % overdue.
Pending Workload Metrics	Active pending count per user, oldest pending age, and pending aging buckets (0-6h, 6-12h, 12-24h, >24h).

15.3 Reporting Architecture

- Aggregations shall be computed asynchronously via Celery and stored in PostgreSQL materialized views or summary tables refreshed at the cadences shown.
- All metrics shall be filterable by date range, account, category, and user (where authorized).
- All charts shall offer CSV export of underlying data (FR-72).

16. Security Requirements

16.1 Security Controls

Control	Implementation
Role-Based Access Control	Centralized policy module checked on every API call; UI hides unauthorized actions.
Password Hashing	Argon2id (preferred) or bcrypt (cost ≥ 12); never stored in plain text.
Secure Session Management	HTTP-only, Secure, SameSite=Strict cookies, or short-lived JWT + refresh token rotation.
Encrypted Credential Storage	eBay credentials, OAuth and refresh tokens encrypted with AES-256-GCM at rest.
Secure API Communication	TLS 1.2+ for all internal and external traffic; HSTS enabled.
CSRF Protection	Double-submit cookie or synchronizer token on every state-changing endpoint.
Rate Limiting	Per-IP and per-user limits on /auth/login, /auth/reset, /conversations/{id}/replies.
Input Validation	Pydantic models on every endpoint; type coercion and bounded string lengths.
Output Encoding	All user-supplied content rendered with React-safe escaping; HTML sanitization for templates.
Data Encryption at Rest	Database disk encryption (LUKS/cloud-managed) plus column-level encryption for tokens.
Data Encryption in Transit	TLS for client-server, backend-database, backend-LLM, and backend-eBay calls.
Audit Logging	Append-only logs per Section 16.2; tamper-evident storage via WORM-style archive.

16.2 Audit Log Categories

Category	Events
User Management	User Created, User Updated, User Deleted, Role Changed, Permission Changed.
Conversation	Assignment, Reassignment, Category Change, Status Change, Spam Action, Resolution, Reopen.

Message	Reply Sent, Attachment Downloaded.
Authentication	Login, Logout, Failed Login.
System	eBay Account Added, eBay Account Removed, Configuration Changes.

16.3 Audit Entry Fields

- Actor User ID and username
- Action code (machine-readable) and human-readable label
- Timestamp (UTC, ISO-8601 with milliseconds)
- IP address (IPv4/IPv6)
- Geographic location (country, region, city — resolved via GeoIP at write time)
- Affected entity type and ID
- Previous value (JSON, redacted for sensitive fields)
- New value (JSON, redacted for sensitive fields)
- Correlation/request ID for tracing

17. Future Enhancements

The following capabilities are explicitly out of scope for v1.0 but are anticipated and should influence architectural decisions to avoid expensive rework.

- Multi-marketplace support (Amazon, Walmart, Shopify, eBay Stores in additional regions). The 'Source' column and adapter pattern in the sync layer should be designed accordingly.
- Escalation engine and email/SMS notifications for SLA breaches.
- Mention support inside the System (e.g., @colleague in internal notes).
- Internal notes / private comments on a conversation (not sent to customer).
- Bulk reply with template macros.
- AI-assisted reply suggestions and auto-drafts.
- Advanced search with saved filters and shared views.
- Customer-facing self-service portal and knowledge base integration.
- Public REST/GraphQL API for third-party integrations.
- Cold-storage archival of attachments older than 12 months (e.g., move from local disk/S3-standard to S3-Glacier or equivalent) with on-demand restore.
- Per-attachment deduplication via content hash to reduce storage.
- Image format conversion (e.g., HEIC → JPEG) on ingestion for broader compatibility.
- Multi-language UI (i18n).
- Mobile-optimized layouts or native applications.

18. Assumptions

ID	Assumption
A-01	eBay's Trading/Messaging APIs remain available and provide the necessary scopes for reading inbox messages, sending replies, and downloading attachments.
A-02	The organization will obtain valid eBay developer credentials for each seller account and supply OAuth refresh tokens during onboarding.
A-03	An LLM endpoint (managed or self-hosted) is available for categorization; cost per classification is acceptable to the business.
A-04	The platform is deployed inside the organization's private network or behind authenticated VPN/SSO; no public internet exposure is required.
A-05	Initial user base is small to medium (≤ 25 concurrent users) consistent with the message volume of 300–400/day.
A-06	All staff use modern desktop browsers; mobile usage is incidental and not a primary use case in v1.0.
A-07	GeoIP resolution will be performed via a licensed local database (e.g., MaxMind GeoLite2) rather than a

	paid online API on every call.
A-08	Attachment sizes follow eBay's existing platform limits; no separate enforcement is required client-side beyond eBay's constraints.
A-09	The 24-hour SLA is uniform; no per-account or per-category SLA variation is required in v1.0.
A-10	Backups, monitoring, and infrastructure operations are owned by the organization's DevOps function and are out of scope for this SRS.

19. Constraints

ID	Constraint
C-01	Technology stack is fixed: React + TypeScript (frontend), FastAPI (backend), PostgreSQL (database), Celery + Redis (background processing), Alembic (migrations).
C-02	The eBay API rate limits and quota policies are external constraints; the System must operate within them via caching, batching, and local persistence.
C-03	Outbound messages must be routed through the eBay Messaging API; no alternative delivery channel exists.
C-04	The platform is for internal use only; customer-facing access is prohibited.
C-05	Notifications in v1.0 are limited to in-app only; integrating email/SMS providers would require additional scope and budget.
C-06	Escalation logic is excluded from v1.0; SLA breach handling is visual only.
C-07	Templates are global; per-user template personalization is excluded from v1.0.
C-08	All times stored in UTC; all displayed times use the viewer's local time zone (BR-15).
C-09	The System must be deployable using container orchestration without proprietary cloud-vendor lock-in.

20. Acceptance Criteria

The System shall be considered acceptable for production release when ALL of the following criteria are demonstrably met in a User Acceptance Testing (UAT) environment with production-equivalent data volume.

ID	Acceptance Criterion
AC-01	All six pre-configured eBay accounts (Main, Marine, Tech, Ship, Tools, Trade) can be connected, synchronized, and replied to end-to-end.
AC-02	A new eBay account can be added via the Admin UI without any code change or redeploy.
AC-03	Inbound messages from a buyer with the same username are grouped into a single conversation per account (BR-01).
AC-04	AI categorization assigns one of the six initial categories with $\geq 85\%$ accuracy on a labeled validation set of 200 messages; low-confidence items route to the Unassigned Queue.
AC-05	An Operation Manager can assign, reassign, and bulk-assign conversations and the receiving Users get an in-app notification within 5 seconds.
AC-06	A User replying from the platform produces a corresponding message visible in the eBay seller back office within 30 seconds.
AC-07	RBAC is enforced: Users cannot access categories not assigned to them, cannot manage users, cannot create templates, and cannot view audit logs.
AC-08	Audit logs contain entries for every action listed in §16.2 with all fields populated per FR-75.
AC-09	The Conversation List view loads ≤ 2 s at the 95th percentile under 25 concurrent users and 250,000 historical messages (NFR-01).
AC-10	Search across customer username and message content (including embedded order numbers) returns correct results within 2 s (FR-62, NFR-02).
AC-11	SLA indicators render correctly across all four buckets (Green/Yellow/Red/Overdue) for the same conversation as time progresses.
AC-12	Notifications appear within 5 seconds of the triggering event and persist for at least 90 days.
AC-13	Templates can be created/edited/deleted only by Admins and are usable by all roles.
AC-14	Attachments in JPG/JPEG/PNG/GIF/WEBP/HEIC/BMP can be viewed inline and downloaded; downloads are audited (FR-57).
AC-15	All eBay credentials are encrypted at rest; no plain-text credentials appear in database dumps or logs (NFR-15).

AC-16	User and Manager analytics widgets display correct counts versus a hand-computed reference dataset, and CSV export matches the displayed data exactly.
AC-17	Daily backups are produced and a sample restore is successfully validated by QA.
AC-18	Penetration test against the OWASP Top 10 is performed; no Critical or High findings remain unmitigated at release.
AC-19	The deployment runs successfully on a single Docker Compose host (staging) and on Kubernetes via Helm (production) using the same image artifacts.
AC-20	Documentation handed to operations includes runbooks for eBay token refresh failures, sync errors, and disaster recovery.

21. Appendix

21.1 Suggested High-Level Data Model

The following entities are suggested as the canonical data model. Field lists are illustrative; the implementation team is expected to refine attributes and add indexes.

Entity	Key Fields	Notes
User	id, username, email, password_hash, role, status, created_at, last_login_at	Role ∈ {Admin, Manager, User}.
Role	id, name, description	Optional; may be implemented as enum on User.
CategoryAccess	user_id, category_id	Many-to-many between User and Category.
EbayAccount	id, label, app_id_enc, cert_id_enc, dev_id_enc, oauth_token_enc, refresh_token_enc, token_expiry, status, last_sync_at, last_sync_error	All *_enc fields are AES-256-GCM.
Category	id, name, is_active, created_by, created_at	Soft-deletable.
Conversation	id, ebay_account_id, buyer_username, subject, status, category_id, assigned_user_id, is_starred_by (m2m), sla_due_at, last_activity_at, created_at	Unique (ebay_account_id, buyer_username).
Message	id, conversation_id, direction (in/out), ebay_message_id, body, sender, sent_at, dispatch_status (for outbound)	Immutable after dispatch.
Attachment	id, message_id, filename, mime_type, size_bytes, storage_path, sha256	sha256 enables future dedup.
Template	id, title, body, category_tag, created_by, updated_at	Globally visible.
Notification	id, user_id, type, conversation_id, message, is_read, created_at	Retain ≥90 days.
AuditLog	id, actor_user_id, action, entity_type, entity_id, prev_value, new_value, ip_address, geo_country, geo_city, occurred_at, request_id	Append-only.
AssignmentHistory	id, conversation_id, from_user_id, to_user_id, by_user_id, occurred_at	Powers reassignment audit and analytics.

21.2 Suggested API Surface (Illustrative)

Method	Endpoint	Purpose
POST	/auth/login	Authenticate user; returns session token.
POST	/auth/logout	Invalidate session.
GET	/users	List users (Admin/Manager).
POST	/users	Create user (Admin).
PATCH	/users/{id}	Edit user.
POST	/users/{id}/disable	Disable user.
DELETE	/users/{id}	Delete user (after reassignment).
GET	/ebay-accounts	List eBay accounts and status.
POST	/ebay-accounts	Add new eBay account (Admin).
POST	/ebay-accounts/{id}/sync	Trigger manual sync.
GET	/conversations	List with filters (account, status, category, assignee, SLA, search).
GET	/conversations/{id}	Detail view including messages and attachments.
POST	/conversations/{id}/assign	Assign or reassign.
POST	/conversations/{id}/status	Change status.
POST	/conversations/{id}/category	Override category.
POST	/conversations/{id}/replies	Send reply.
GET	/attachments/{id}/download	Download attachment (audited).
GET	/templates	List templates.
POST	/templates	Create template (Admin).
GET	/analytics/user	Personal analytics.
GET	/analytics/manager	Team analytics (Manager/Admin).
GET	/audit	Filtered audit log query (Admin).
GET	/audit/export	CSV export of audit log.
GET	/notifications	List notifications.

POST	/notifications/{id}/read	Mark as read.
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21.3 Recommended Storage Optimization (Attachments)

- Persist attachments to object storage (S3-compatible) rather than the local file system to enable horizontal scaling.
- Compute SHA-256 on ingestion and deduplicate identical attachments across conversations.
- After 12 months of inactivity, transition attachments to a cold-storage tier (e.g., S3 Glacier or Azure Archive).
- Convert HEIC attachments to JPEG on ingestion for broader viewer compatibility while preserving the original.
- Track storage size per eBay account to inform cost allocation and quota policies.

21.4 Recommended Background Jobs

Job	Schedule	Purpose
sync_account	Every 2 minutes per account	Fetch new messages from eBay.
dispatch_reply	On-demand	Send outbound replies to eBay with retry.
categorize_conversation	On new message	Call LLM classifier and persist category + confidence.
refresh_ebay_tokens	Daily / pre-expiry	Refresh OAuth tokens before they expire.
rebuild_analytics	Hourly	Refresh materialized views / summary tables for analytics.
geoip_resolve	On audit write	Resolve IP to country/city at write time.
backup_database	Daily	Logical and/or physical backup with off-site copy.
archive_old_attachments	Weekly	Move attachments >12 months to cold storage.

21.5 Traceability Matrix (Sample)

The development team shall maintain a full traceability matrix mapping each requirement (FR/NFR/BR) to design artifacts, code modules, and test cases. A sample fragment is provided below.

Requirement	Design Artifact	Test Case
FR-27 (One conversation per buyer/account)	Conversation entity + unique index	TC-CONV-001
FR-58 (SLA indicators)	SLAService + UI badge component	TC-SLA-001..004
FR-62 (Search)	PostgreSQL FTS + search service	TC-SEARCH-001..006

NFR-15 (Encryption at rest)	Token encryption service + KMS	TC-SEC-002
BR-03 (Auto-reopen)	Inbound message handler	TC-CONV-014
FR-74..78 (Audit)	AuditLog entity + middleware	TC-AUDIT-001..010

21.6 Glossary of Statuses & Colors

Status / Indicator	Meaning	Color
Open	Awaiting action from support staff.	Neutral / Blue
Pending	Awaiting information from customer; SLA paused.	Amber
Resolved	Issue completed; closed.	Green
Reopened	Customer replied after resolution; treated as Open.	Blue
Spam	Irrelevant or unwanted communication.	Grey
SLA Healthy	>50% time remaining.	Green
SLA Warning	≤50% time remaining.	Yellow
SLA Critical	≤10% time remaining.	Red
SLA Overdue	Exceeded 24-hour SLA.	Dark Red / Black border

— End of Software Requirements Specification —